

TIC 392559026

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 2.671276 d
Epoch (t0) 2308.7640 BTJD
Duration 0.0840 d
Depth 4.302 % (43,025 ppm)
SNR 90.1

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 48 (gate ≥ 3)
Robust depth ratio 1.05 (gate ≥ 0.50)
Robust depth sigma 75.8 (gate ≥ 7.0)
Odd/even z-score 2.97 (consistent)
Secondary eclipse detected, phase 0.49, 5.7 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 64 light-curve dip in this sector: 38.4 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 30.60 (gate ≥ 5.0) Compactness 0.85 (gate ≥ 0.50)
Dimming centroid offset 0.13 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 392559018
0.58 px away, dTmag +1.37 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

